TANMOY PAL

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WORK EXPERIENCE

Senior Manager, TATA Motors, Jamshedpur, India — 2023 - Present

TPM - Digital Transformation and Process automation

AI-based Prescriptive System with Spare Parts Management Portal - Led the product strategy for a multi-agent RAG system, prioritizing feature roadmaps for web search and vector database agents to automate breakdown resolution, reduced MTTR by 10%.

Factory Digital Twin - Orchestrated cross-functional delivery between OT and IoT teams to integrate real-time edge data into a Framence and Dassault 3DEXPERIENCE platform, enabled real-time operational visibility, eliminated manual reporting.

Autonomous Mobile Robot - Managed the end-to-end product lifecycle of an in-house AMR development, aligning engineering and supply chain, and line production stakeholders to drive a 26% efficiency gain.

Switch Tray verification system with Cab verifier - Spearheaded the development of a CV-based quality product, managing stakeholder requirements to eliminate manual checks and saved INR 20 million.

Digital Kitting Dashboard (Pick-to-Light)-Led the feature mapping and implementation for a smart kitting system, eliminating part mismatch errors and reducing line-side material search time by 70%, improving first-time-right assembly rates by 70%.

Breakdown Management Portal - Prioritized the development of IoT-driven predictive models and conveyor current pattern analytics to transition factory maintenance from reactive to a proactive model, reducing unplanned downtime by 22%.

Digital Andon System - Implemented digital ANDON system for line efficiency calculations, automating real-time data ingestion from OT layer and improving accuracy by 100% in the dashboard with operational KPIs**,**eliminating manual MS-excel tracking.

DISHA Manpower Allocation Portal - Built and launched a workforce allocation tool, balancing stakeholder needs with complex shop-floor scheduling and skill-level mapping portal for daily work management according to factory production line operation and work scheduling. Improved workforce utilization by 15% and reduced allocation conflicts.

Manufacturing Part Status System - Implemented and defined the product roadmap for an inventory-based scheduling system, solving critical item-tracking pain points through real-time data visibility. It reduced part-tracking discrepancies by 28%.

Software Engineer, SIEMENS, Bangalore, India — 2022 – 2023

TOTALLY INTEGRATED AUTOMATION PORTAL

Implemented various product backlog items as features for import/export functionality in the TIA Portal, including tool-based integration with MDD and GSD tools.

Developed multiple POCs to enhance product functionality and automate processes for improved efficiency.

Developed features on export functionalities of new devices in TIA.

Implemented test automations on the newly built features **.**

Built and validated an XML parsing framework reducing configuration error by 20%.

R&D Member of Technical Staff I, AVEVA(Schneider Electric), Hyderabad, India — 2019 - 2022

AVEVA ENGINEERING AND SCHEMATICS -

Applied ADS in UI and product design using WPF, adhering to new design standards for enhanced user experience.

Automated the feature test cases which reduced manual QA effort by 5% of the total automation .

AVEVA STRUCTURAL DESIGN -

Developed a graphics feature with an algorithm based on geometrical properties for relative beams and columns, reducing 400 lines of code and doubling efficiency.

Implemented product integration with other products with Structural 3D/BOCAD by the creation of suitable API layer features.

Automated manual workflows using desktop-based automation methods, which increased the product test case efficiency by 16%.

Implemented product integration framework of Aveva structural design which can interface with different softwares.

EDUCATION

Bachelor of Technology – Computer Science and Engineering — **2015 - 2019**

KIIT University, Bhubaneswar, India

EXECUTIVE PROGRAM IN MANAGEMENT & BUSINESS ADMINISTRATION — **2024 - 2025**

Institute of Management, Technology and Finance, Lisbon, Portugal

Product management fellowship – Product Space 2024 **McKinsey Forward Program -McKinsey & Company** 2025

Product management certificate-Institute of Product Leadership **Lean Six Sigma Black Belt - SSAA** 2025

SKILLS- SDLC(C#,Python, React),Agile development ,Stakeholder management ,Agentic AI-(Langchain ,Autogen,ADK), Product prototyping,Analytics,Power BI ,Industrial digital product development,IIoT ,Hardware product development–(Robotics and ROS2),NPI , Product strategy,SQL,Data Visualization,Deep learning

ACHIEVEMENTS- **Awarded the Plant's Outstanding Contribution and Organizational Silver Awards** for leading high-impact digital transformation and Autonomous Mobile Robot (AMR) implementations. **Recognized as a Top Product Management Fellow** and multi-award winner at national CII competitions for excellence in AI-driven product development and innovation.